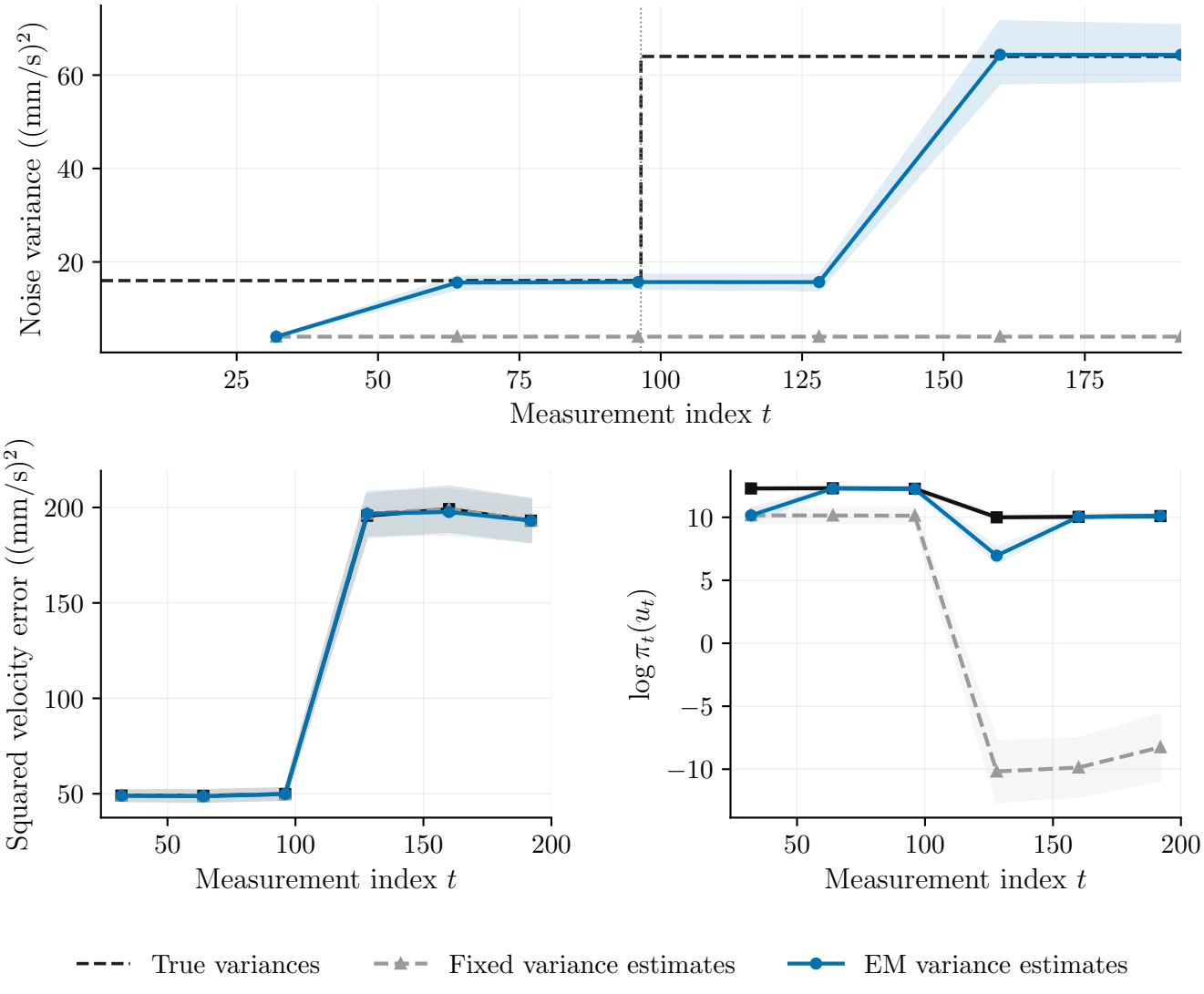
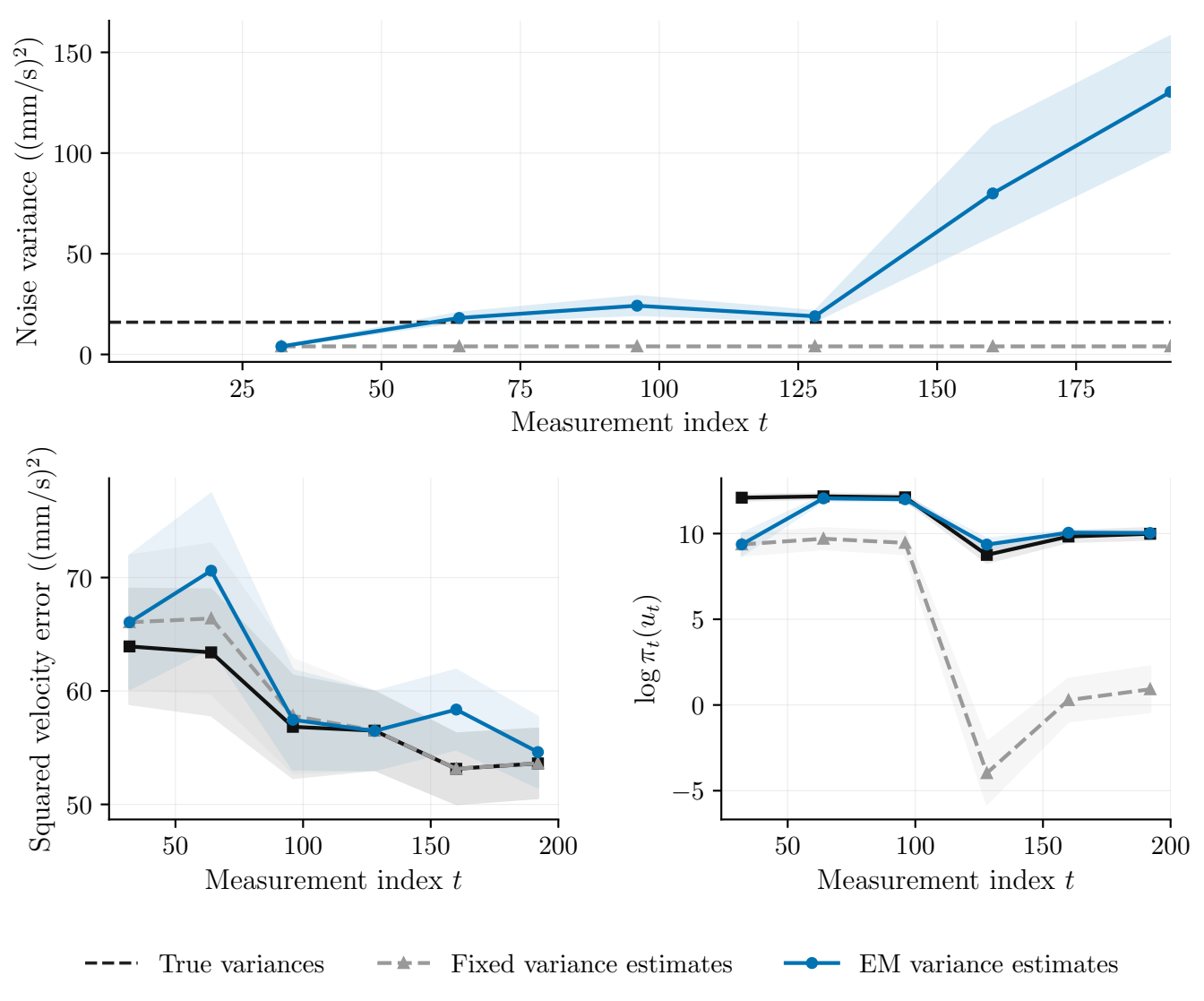
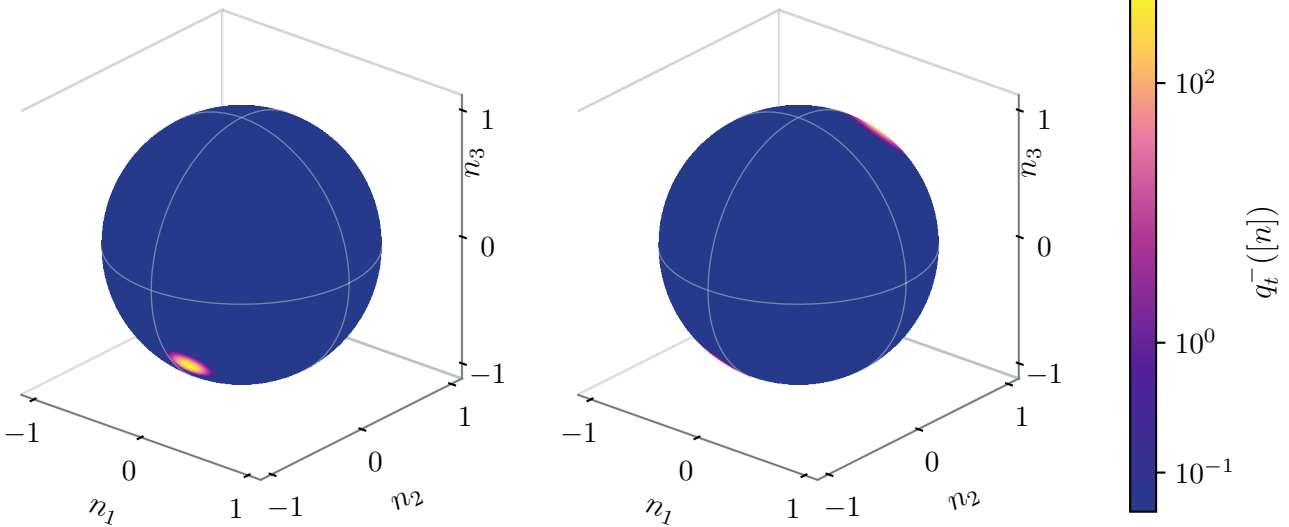
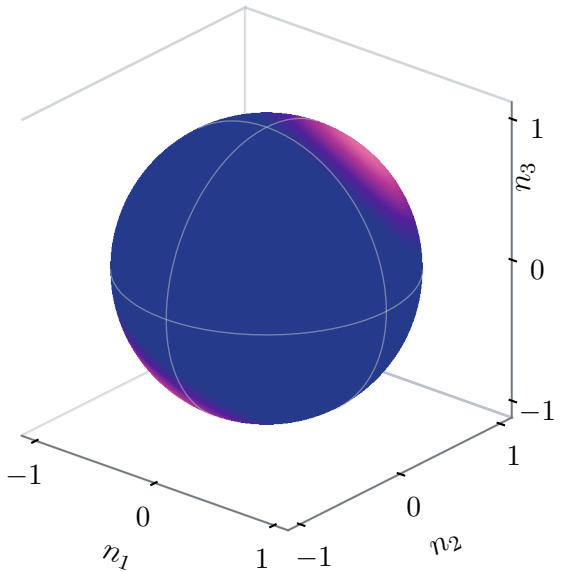
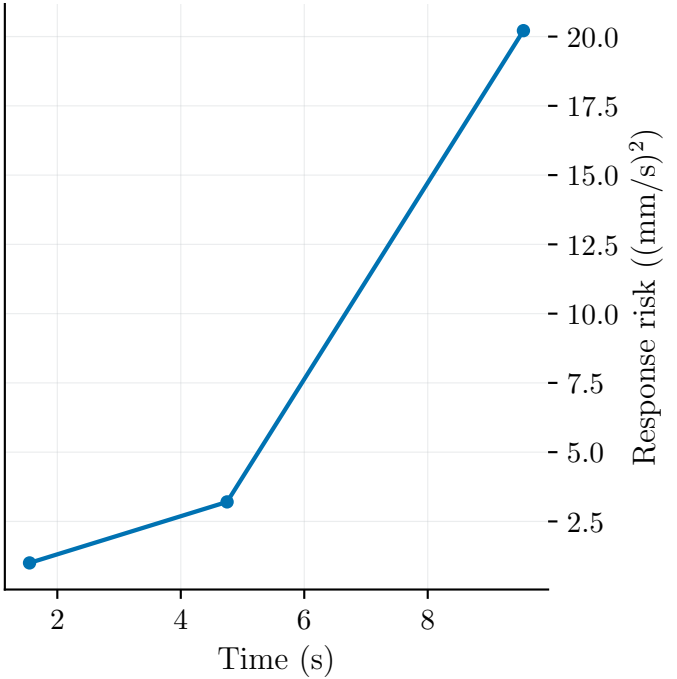
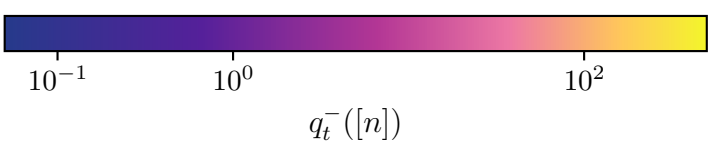

Noise variance ((mm/s)^2)
20
40
60
25
50
75
100
125
150
175
Measurement index t
Squared velocity error ((mm/s)^2)
50
100
150
200
50
100
150
200
Measurement index t
$\log \pi_t(u_t)$
10
5
0
−5
−10
50
100
150
200
Measurement index t
True variances
Fixed variance estimates
EM variance estimates


Noise variance ((mm/s)^2)
150
100
50
0
25
50
75
100
125
150
175
Measurement index t
Squared velocity error ((mm/s)^2)
70
60
50
50
100
150
200
Measurement index t
$\log \pi_t(u_t)$
10
5
0
−5
50
100
150
200
Measurement index t
True variances
Fixed variance estimates
EM variance estimates


1
0
−1
n_3
−1
0
1
n_1
−1
0
1
n_2
10^2
10^0
10^{-1}
$q_t^-([n])$


1
n_3
0
−1
−1
0
1
n_1
1
0
−1
n_2


20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Response risk ((mm/s)^2)
2
4
6
8
Time (s)


10^{-1}
10^{0}
10^{2}
$q_t^-([n])$